

DATA ANALYTICS FINAL PROJECT

AMAZON ORDERS DATASET

INSIGHTS & STRATEGY REPORT

Prepared by

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November 2025 – April 2026

Tools Used:

Python | Microsoft Excel | MySQL Workbench | Power BI

Dataset: Amazon Orders | Rows: 1,13,698 | Period: Jan 2022 – Sep 2022

1. Executive Summary

This report presents a comprehensive data-driven analysis of Amazon Orders dataset containing 1,13,698 records spanning January 2022 to September 2022. The analysis was conducted as part of the Data Analytics Final Project at Agileology Institute, utilizing four industry-standard tools — Python, Microsoft Excel, MySQL Workbench, and Power BI.

The primary objective of this project was to explore, clean, and analyze the Amazon Orders dataset to extract meaningful business insights about customer ordering patterns, revenue trends, product performance, and delivery operations. The findings from this analysis are intended to assist Amazon sellers in making data-driven decisions to improve sales performance and operational efficiency.

Key Performance Indicators at a Glance

1,13,698 Total Orders	INR 75M+ Total Revenue	INR 663 Avg Order Value	1,14,136 Total Qty Sold
6,000+ Cancelled Orders	4.95% Cancellation Rate	Kurta / Set Top Category	Maharashtra Top State

Project Scope

The analysis covered the following key areas:

- Data Cleaning and Preprocessing using Python (Pandas, NumPy, Matplotlib, Seaborn)
- Exploratory Data Analysis with pivot tables, charts and dashboards in Microsoft Excel
- Structured Query Language analysis using 20 business queries in MySQL Workbench
- Interactive 5-page Dashboard built in Microsoft Power BI Desktop
- Insights extraction and business strategy recommendations based on findings

Dataset Overview

Attribute	Details
Dataset Name	Amazon Orders Dataset
Total Records (After Cleaning)	1,13,698 rows
Total Columns	29 columns

Attribute	Details
Time Period	January 2022 to September 2022
Geography	India (Pan-India Orders)
Source Type	E-Commerce Transaction Data
Key Columns	Order ID, Date, Status, Category, Size, Amount, Qty, City, State

2. Data-Driven Observations

The following observations are derived directly from the cleaned dataset through Python EDA, Excel Pivot Analysis, MySQL Queries, and Power BI Dashboard visuals.

2.1 Order Status Distribution

Order Status	Order Count	Percentage (%)
Shipped	76,000+	66.91%
Shipped - Delivered To Buyer	28,000+	24.60%
Cancelled	6,000+	4.95%
Shipped - Returned To Seller	~2,000	~1.80%
Pending	~500	~0.40%

KEY

Over 91% of orders were successfully shipped or delivered, indicating strong fulfilment performance. The 4.95% cancellation rate is within acceptable industry standards for Indian e-commerce.

2.2 Product Category Analysis

The dataset covers 9 product categories — all belonging to the Indian ethnic and western fashion segment. The top performing categories by order volume were:

Rank	Category	Total Orders	Share (%)
1	Kurta	44,000+	38.76%
2	Set	44,000+	38.61%
3	Western Dress	14,000+	12.04%
4	Top	10,000+	8.52%
5	Ethnic Dress	~2,000	~1.80%
6	Blouse / Bottom / Saree / Dupatta	< 1,000 each	< 1% each

KEY

Kurta and Set together dominate 77% of all orders — clearly the two highest-demand categories on this Amazon seller account. Western Dress at 12% shows growing demand for western wear.

2.3 Product Size Analysis

Size distribution reveals strong preference for standard sizes. Based on the analysis:

- M (Medium) and L (Large) are consistently the most ordered sizes across all categories
- XL and XXL show significant demand — indicating a strong plus-size customer base
- XS and Free sizes have the least orders — suggesting niche demand
- 3XL, 4XL, 5XL, 6XL together account for a small but growing segment

KEY

Sellers should maintain higher inventory for M, L, XL, and XXL sizes. XS and Free sizes can be stocked in limited quantities to avoid unsold inventory.

2.4 Geographic Distribution

Top 5 States by Revenue

Rank	State	Total Revenue (INR)	Orders
1	Maharashtra	12.9 Million	Highest
2	Karnataka	10.2 Million	2nd
3	Telangana	6.6 Million	3rd
4	Uttar Pradesh	6.5 Million	4th
5	Tamil Nadu	6.2 Million	5th

Top 5 Cities by Revenue

Rank	City	Total Revenue (INR)
1	Bengaluru	7.1 Million
2	Hyderabad	5.4 Million
3	Mumbai	4.2 Million
4	New Delhi	3.8 Million
5	Chennai	3.5 Million

KEY

The top 5 states — Maharashtra, Karnataka, Telangana, UP, and Tamil Nadu — together contribute over 55% of total revenue. Metro cities like Bengaluru, Hyderabad, and Mumbai drive the highest city-level revenue.

3. Trends in Customer Orders

3.1 Monthly Revenue Trend

The monthly revenue analysis revealed a highly seasonal pattern across the dataset period:

Month	Total Revenue (INR)	Trend
January	2.4 Million	Low
February	2.5 Million	Low
March	2.9 Million	Growing
April	2.7 Million	Stable
May	17.1 Million	PEAK
June	14.7 Million	High
July	2.7 Million	Drop
August	2.4 Million	Low
September	19.6 Million	PEAK
October	2.6 Million	Drop
November	2.9 Million	Growing
December	2.8 Million	Stable

PEAK

May (INR 17.1M) and September (INR 19.6M) are the two highest revenue months. September is the single biggest month — likely driven by festive season demand (Navratri, Durga Puja, Dussehra). May peak likely driven by summer fashion buying.

3.2 Fulfilment Channel Trends

Fulfilment Type	Total Orders	Total Revenue	Avg Order Value
Amazon (FBA)	Majority	Higher Share	Higher
Merchant (FBM)	Minority	Lower Share	Slightly Lower

KEY

Amazon-fulfilled orders significantly outnumber merchant-fulfilled orders, confirming that FBA (Fulfilled by Amazon) is the dominant and preferred fulfilment method on this account.

3.3 Shipping Service Level Trends

Service Level	Total Revenue (INR)	Observation
Expedited	17.8 Million	Dominant preference — customers prefer faster delivery
Standard	6.1 Million	Smaller share — price-sensitive customers

KEY	Expedited shipping accounts for nearly 75% of total revenue. This shows that Amazon customers, especially in metro cities, strongly prefer faster delivery even if it costs more.
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3.4 B2B vs B2C Trends

Customer Type	Orders	Percentage
B2C (Individual Customers)	1,13,000+	99.27%
B2B (Business Customers)	~1,000	0.73%

KEY	This is overwhelmingly a B2C business — 99.27% individual customers. B2B orders are negligible at 0.73%, indicating the seller primarily caters to retail fashion consumers.
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4. Factors Influencing Higher Order Amounts

Based on the SQL analysis and Power BI dashboard findings, the following factors were identified as key drivers of higher order amounts:

4.1 Promotions Drive Revenue Significantly

Promotion Type	Total Revenue (INR)	Avg Order Value
Promoted Orders	54 Million	Higher
No Promotion	22 Million	Lower

FINDING

Promoted orders generate 2.5x more revenue than non-promoted orders. This strongly suggests that promotions and discount campaigns are a critical driver of sales volume on Amazon.

4.2 Expedited Shipping Correlates with Higher Spend

- Expedited shipping customers spend significantly more per order than standard shipping customers
- This suggests higher-value product buyers prefer faster delivery — possibly premium customers
- Standard shipping orders tend to be lower value, price-sensitive purchases

FINDING

There is a clear positive correlation between shipping speed preference and order value. Sellers targeting premium buyers should ensure Expedited shipping availability at all times.

4.3 Category Influences Order Value

Category	Avg Order Value	Revenue Rank
Set	Highest	1st
Western Dress	High	2nd
Kurta	Medium-High	3rd
Ethnic Dress	Medium	4th
Top / Blouse / Bottom	Lower	5th-7th

FINDING

Sets and Western Dresses command higher average order values compared to individual items like Kurtas and Tops. Bundling products as Sets is an effective strategy to increase order value.

4.4 Geography Influences Order Amount

- Metro cities (Bengaluru, Hyderabad, Mumbai, Delhi, Chennai) consistently show higher average order values
- Tier-2 and Tier-3 cities show lower average order values but higher order volumes in some states
- Maharashtra and Karnataka lead in both volume and revenue — dual strength markets

FINDING

Metro cities are premium markets with higher spending per order. Tier-2 cities like Jaipur, Pune, and Ahmedabad are volume markets with growth potential.

5. Business Strategies & Recommendations

Based on the data analysis findings, the following strategies are recommended for Amazon sellers in the Indian ethnic and western fashion segment:

5.1 Inventory & Product Strategy

- Focus heavily on Kurta and Set categories — they account for 77% of all orders. Always maintain adequate stock for these two categories.
- Stock sizes M, L, XL and XXL in higher quantities. These are consistently the best-selling sizes across all categories.
- Consider launching more bundled Set products as they command higher average order values than individual items.
- Reduce slow-moving SKUs in Dupatta, Blouse, and Bottom categories — they show very low order volumes.
- Expand the 3XL and 4XL size range — plus-size demand is growing and underserved.

5.2 Geographical Targeting Strategy

- Prioritize marketing and ad spend in Maharashtra and Karnataka — these are the highest revenue states.
- Focus on Bengaluru, Hyderabad, and Mumbai for premium product launches — these cities show the highest per-order spending.
- Expand aggressively into Uttar Pradesh and Tamil Nadu — they are in the Top 5 revenue states and show strong growth potential.
- Consider regional sizing preferences — northern states may prefer different fits compared to southern states.

5.3 Seasonal & Timing Strategy

- Maximise inventory and run aggressive promotions in September — it is the single highest revenue month, driven by festive season (Navratri, Durga Puja, Dussehra).
- Prepare inventory in advance for May sales surge — likely driven by summer collection demand.
- Plan clearance sales in low-revenue months (January, February, August) to clear unsold inventory.
- Schedule new product launches in March-April to capitalise on pre-summer demand growth.

5.4 Promotion & Pricing Strategy

- Run promotions consistently — promoted orders generate 2.5x more revenue than non-promoted orders. This is the single most impactful lever.

- Use Amazon's Lightning Deals and Coupons during festive seasons to maximise visibility and conversion.
- Offer free expedited shipping for orders above INR 800 to encourage larger basket sizes.
- Price Sets between INR 700-1000 range — this is the Revenue_Band with highest volume (Premium band).

5.5 Fulfilment & Operations Strategy

- Continue using Amazon FBA (Fulfilled by Amazon) as the primary fulfilment method — it drives majority of revenue and customer satisfaction.
- Ensure Expedited shipping is available for all best-selling SKUs — 75% of revenue comes from expedited orders.
- Monitor Courier Status regularly — any increase in Undelivered or Returned orders needs immediate attention as it affects seller ratings.
- Work on reducing the 4.95% cancellation rate — even reducing it to 3% could recover significant revenue.

6. Recommendations for New Sellers on Amazon India

Based on the patterns observed in this dataset, here are practical recommendations specifically for new sellers entering the Amazon India fashion marketplace:

6.1 Category Selection

- Start with Kurta or Set category — they have the highest proven demand (77% market share in this dataset).
- Avoid starting with low-volume categories like Dupatta, Saree, or Blouse as a new seller — competition is high and volumes are low.
- If starting with Western Dress — target metro cities specifically as that's where demand is concentrated.

6.2 Size Range to Launch With

- Launch with at minimum these 4 sizes: M, L, XL, XXL — they cover approximately 70% of all orders.
- Add XS and 3XL in the second phase once your main sizes are selling well.
- Avoid investing heavily in 5XL and 6XL initially — very low demand currently.

6.3 Pricing Guidance

Product Type	Recommended Price Range	Reasoning
Kurta (Single)	INR 350 – 650	High volume, price-sensitive buyers
Set (2-3 piece)	INR 600 – 1000	Higher value, justifies bundle pricing
Western Dress	INR 500 – 900	Premium urban buyers
Top / Bottom	INR 250 – 500	Affordable, impulse purchase range
Ethnic Dress	INR 500 – 900	Occasion wear, moderate spend

6.4 Operational Recommendations

- Enrol in Amazon FBA from Day 1 — it significantly improves delivery speed and customer satisfaction compared to self-shipping.
- Always offer Expedited shipping — 75% of revenue in this dataset came from expedited orders.
- Run at least one promotion or coupon on your listings from the start — promoted orders generate 2.5x more revenue.
- Focus your marketing budget on Maharashtra, Karnataka, and Uttar Pradesh first — these states give the best return on ad spend.

- Maintain a cancellation rate below 2.5% — Amazon penalises sellers with high cancellation rates affecting account health.
- Monitor your inventory for M and L sizes most carefully — these are most frequently out-of-stock for fashion sellers.

6.5 Growth Timeline Suggestion

Phase	Timeline	Focus Area
Phase 1 — Launch	Month 1-2	List Kurta & Set in M, L, XL, XXL with promotions
Phase 2 — Optimize	Month 3-4	Analyse data, optimise pricing, expand sizes
Phase 3 — Scale	Month 5-6	Add Western Dress, expand to more cities
Phase 4 — Festive Push	August-September	Heavy promotions for festive season peak
Phase 5 — Review	October onwards	Review performance, plan next year strategy

7. Conclusion

This project successfully demonstrated the complete end-to-end Data Analytics workflow applied to a real-world e-commerce dataset. The Amazon Orders dataset of 1,13,698 records was thoroughly cleaned, analysed, and visualised using four industry-standard tools — Python, Microsoft Excel, MySQL Workbench, and Microsoft Power BI.

The analysis revealed that the Indian fashion e-commerce market, as reflected in this dataset, is dominated by ethnic wear categories (Kurta and Set), driven heavily by metro cities, with a strong preference for Amazon-fulfilled and expedited shipping orders. Promotional campaigns are clearly the most impactful lever for revenue growth, generating 2.5 times more revenue compared to non-promoted orders.

The September festive season surge and the May summer peak represent the two most critical selling windows that sellers must prepare for in advance. Geographic concentration in Maharashtra and Karnataka, with emerging growth in Uttar Pradesh and Tamil Nadu, provides a clear roadmap for targeted expansion.

Key Takeaways

- Kurta and Set are the backbone of this seller's business — maintain consistent stock always.
- Metro cities drive premium revenue — Mumbai, Bengaluru, Hyderabad are top priority markets.
- September and May are peak months — plan inventory and promotions 6-8 weeks in advance.
- Promotions are non-negotiable — they are directly responsible for 71% of total revenue.
- Expedited FBA is the winning formula — 75% of revenue comes from Amazon-fulfilled expedited orders.
- Cancellation rate of 4.95% needs attention — even small improvements can recover significant revenue.

Project Deliverables Summary

Tool	Deliverable	Description
Python	Data_Cleaning.py + Cleaned_Amazon_Dataset_Final.csv	Data cleaning, EDA, and 6 chart outputs
Excel	Amazon_Analysis.xlsx	8 pivot tables with charts and KPI dashboard
MySQL	Amazon_Queries.sql	20 business SQL queries with results
Power BI	Amazon_Dashboard.pbix	5-page interactive dashboard
Report	Insights_Strategy_Report.docx	This document — insights and recommendations

End of Report

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